

Chapter 3: Functional group protection

General considerations

- Main features of a protecting group

- Compatibility and orthogonality between protecting groups

Classification of protecting groups according to the cleavage conditions

Functional group protection

- Protection of hydroxyl, carbonyl, carboxyl and amino groups

References:

- ✓ F.A. Carey, R.J. Sundberg, *Advanced Organic Chemistry. Part B: Reactions and Synthesis*, 4^a ed., Kluwer, 2001. Capítol 13.1
- ✓ T.W. Greene, P.G.M. Wuts, *Protective Groups in Organic Synthesis*, 3^a ed., Wiley, 1999
- ✓ P.J. Kociński, *Protecting Groups*, Thieme, 1994.
- ✓ J. Robertson, *Protecting Group Chemistry*, Oxford Chemistry Primers, 2000

Chapter 3. Functional group protection

3.1. General considerations

- Issues that define the suitability of a protecting group:
 - a) It should be cheap or readily available
 - b) It should be easily and efficiently introduced
 - c) It should be easy to characterize and avoid such complications as the creation of new stereogenic centers
 - d) It should be stable to the widest possible range of reaction conditions and to the work-up steps
 - e) It should be stable to the different separation and purification techniques, such as chromatography
 - f) It should be removed selectively and efficiently under highly specific conditions
 - g) The by-products of the deprotection should be easily separated from the substrate

- Two protecting groups are compatible if they show a gradual lability towards a given reagent.

- Two protecting groups are orthogonal if one can be removed in the presence of the other and in any order. Their removal mechanisms are different.

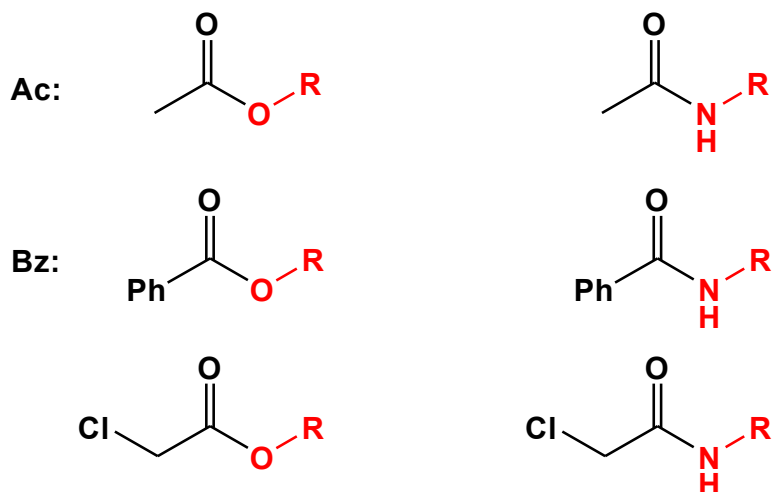
Chapter 3. Functional group protection

3.2. Classification of protecting groups according to the cleavage conditions

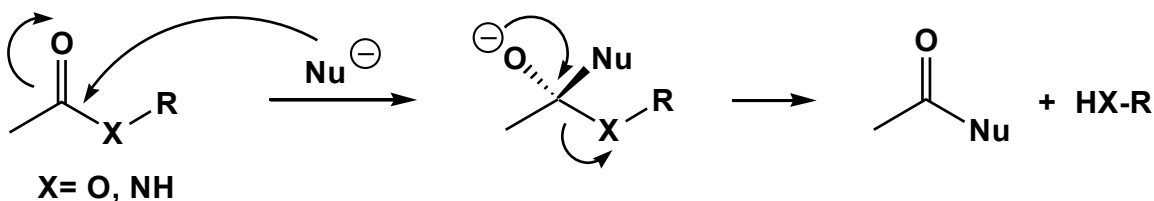
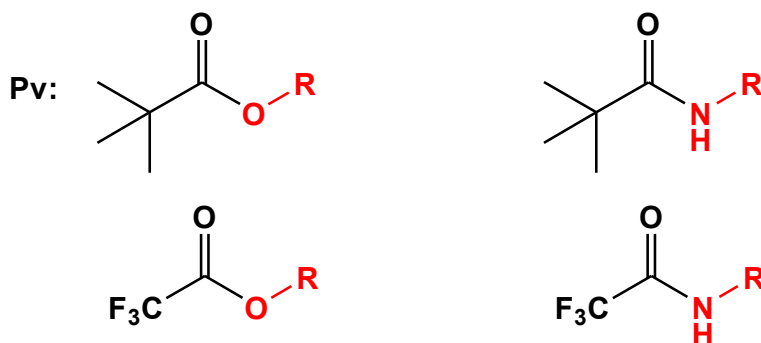
- 3.2.1. Protecting groups cleaved by basic solvolysis
- 3.2.2. Protecting groups cleaved by acid
- 3.2.3. Protecting groups cleaved by a heavy metal
- 3.2.4. Protecting groups cleaved by fluoride ions
- 3.2.5. Protecting groups cleaved by reductive elimination
- 3.2.6. Protecting groups cleaved by β -elimination reactions
- 3.2.7. Protecting groups cleaved by hydrogenolysis
- 3.2.8. Protecting groups cleaved by oxidation
- 3.2.9. Protecting groups cleaved by dissolving metal reduction
- 3.2.10. Protecting groups cleaved by transition metal catalysis
- 3.2.11. Protecting groups cleaved by light

3.2.1. Protecting groups cleaved by basic solvolysis

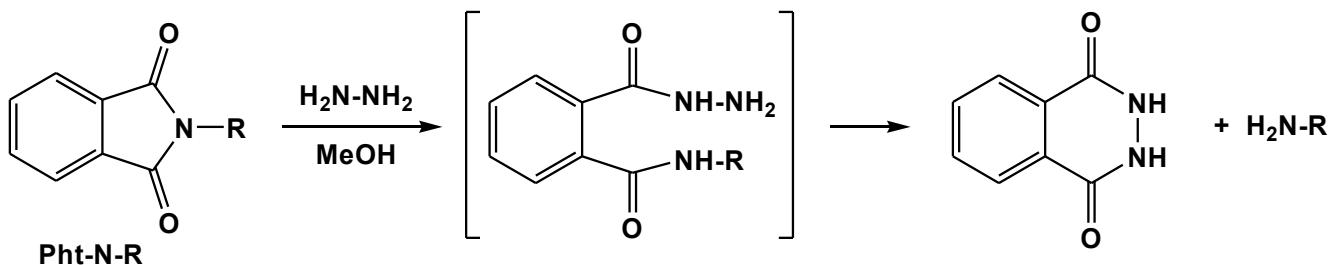
Acyl derivatives of hydroxyl and amino groups:



Chapter 3. Functional group protection



Phthaloyl group, a special case:



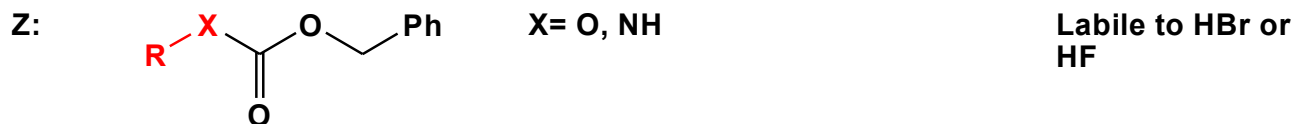
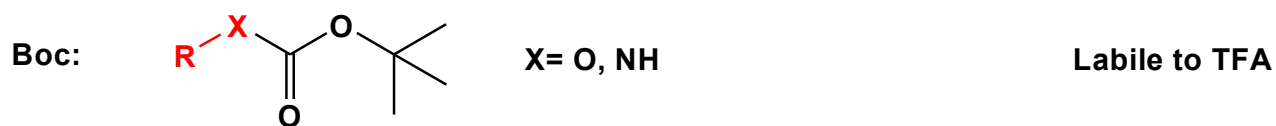
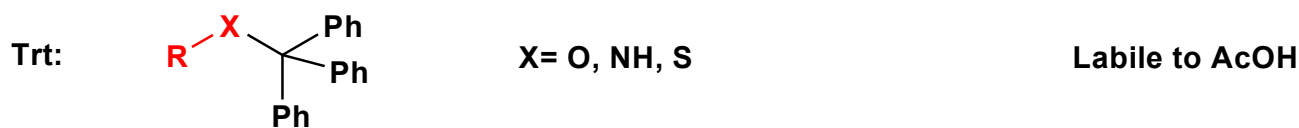
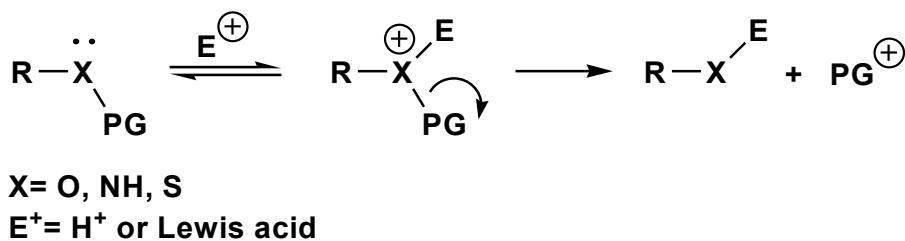
3.2.2. Protecting groups cleaved by acid

They can be divided into two subsets:

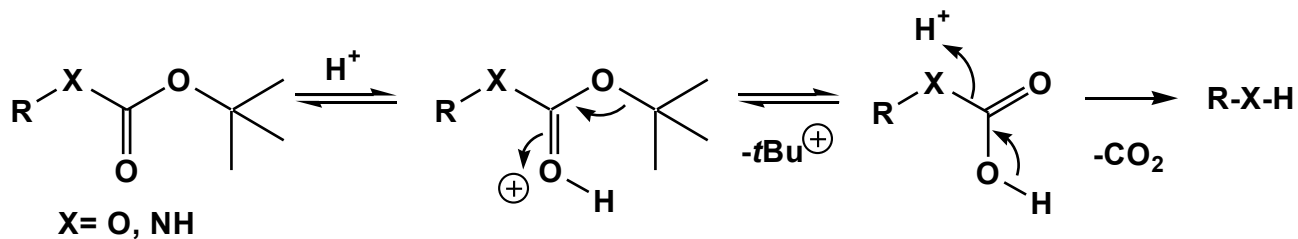
- Tertiary alkyl or benzylic amines, ethers, esters, carbonates and carbamates
- O,O-acetals

Chapter 3. Functional group protection

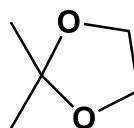
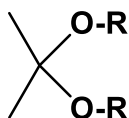
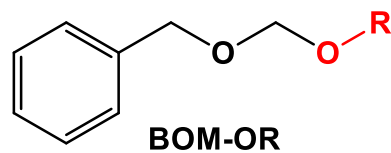
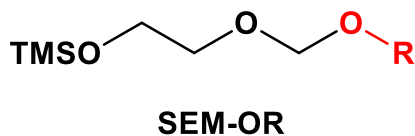
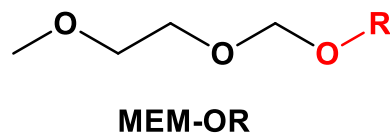
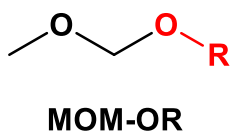
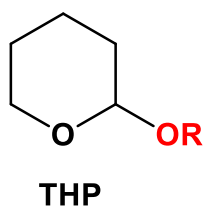
- Tertiary alkyl or benzylic amines, ethers, esters, carbonates and carbamates



Chapter 3. Functional group protection



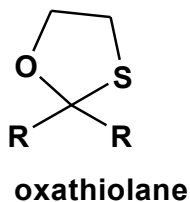
➤ O,O-acetals



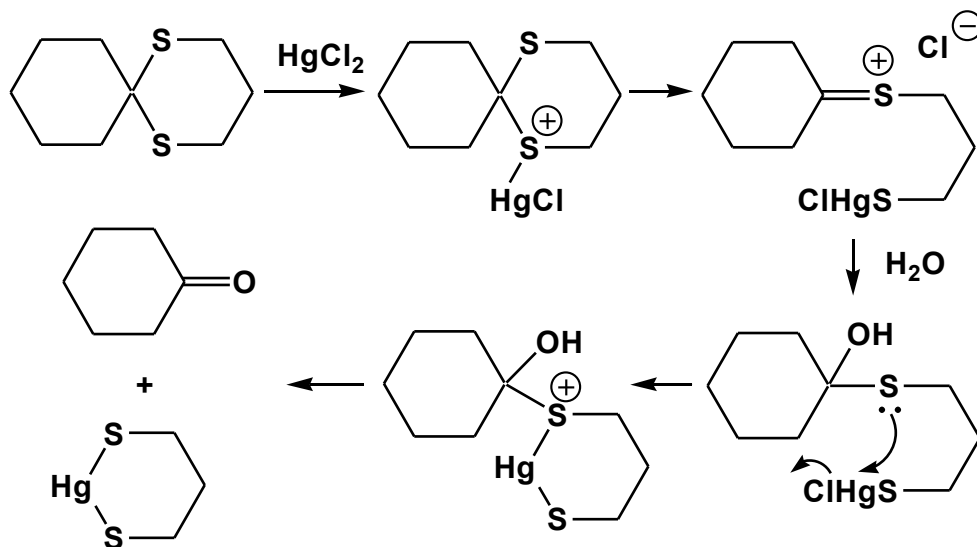
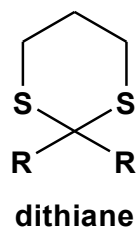
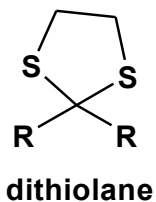
Chapter 3. Functional group protection

3.2.3. Protecting groups cleaved by a heavy metal

O,S-acetals



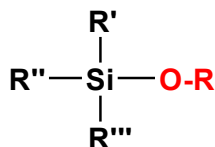
S,S-acetals



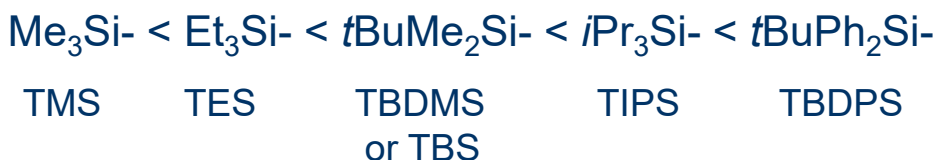
Chapter 3. Functional group protection

3.2.4. Protecting groups cleaved by fluoride ions

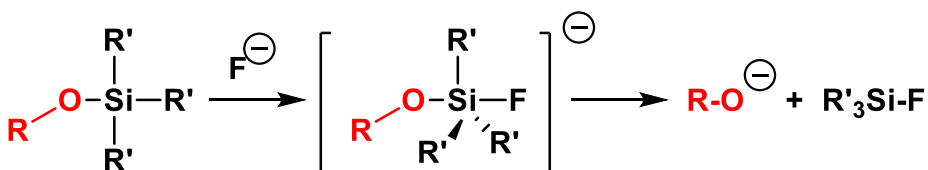
➤ Cleavage of Si-O bonds: silyl ethers



Acid catalyzed
solvolysis:

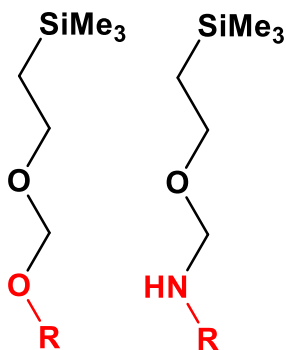
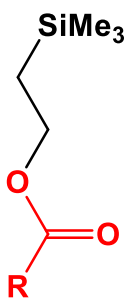


Base catalyzed
solvolysis:



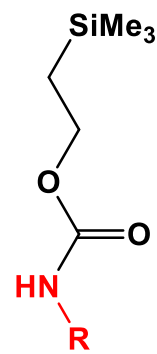
Reagents: $n\text{Bu}_4\text{N}^+\text{F}^-$ (TBAF) in THF or HF in MeCN

➤ Cleavage of Si-C bonds



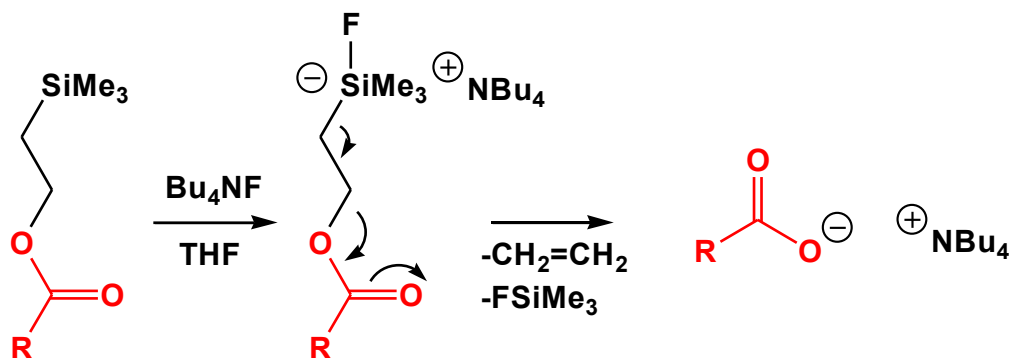
SEM

TMSEC



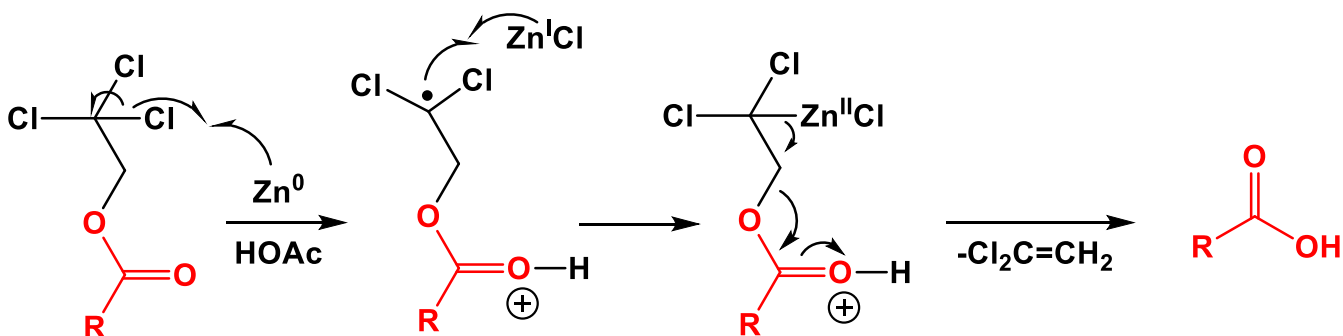
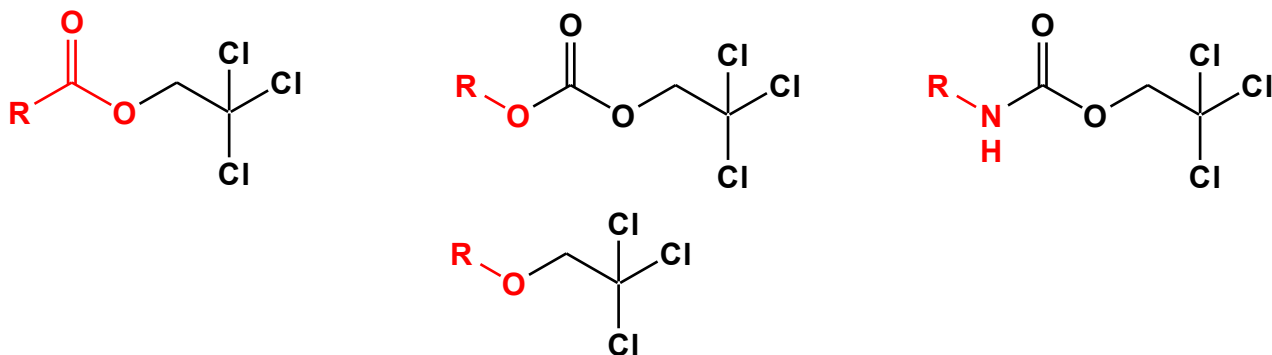
Teoc

Chapter 3. Functional group protection



3.2.5. Protecting groups cleaved by reductive elimination

2,2,2-Trichloroethyl derivatives:

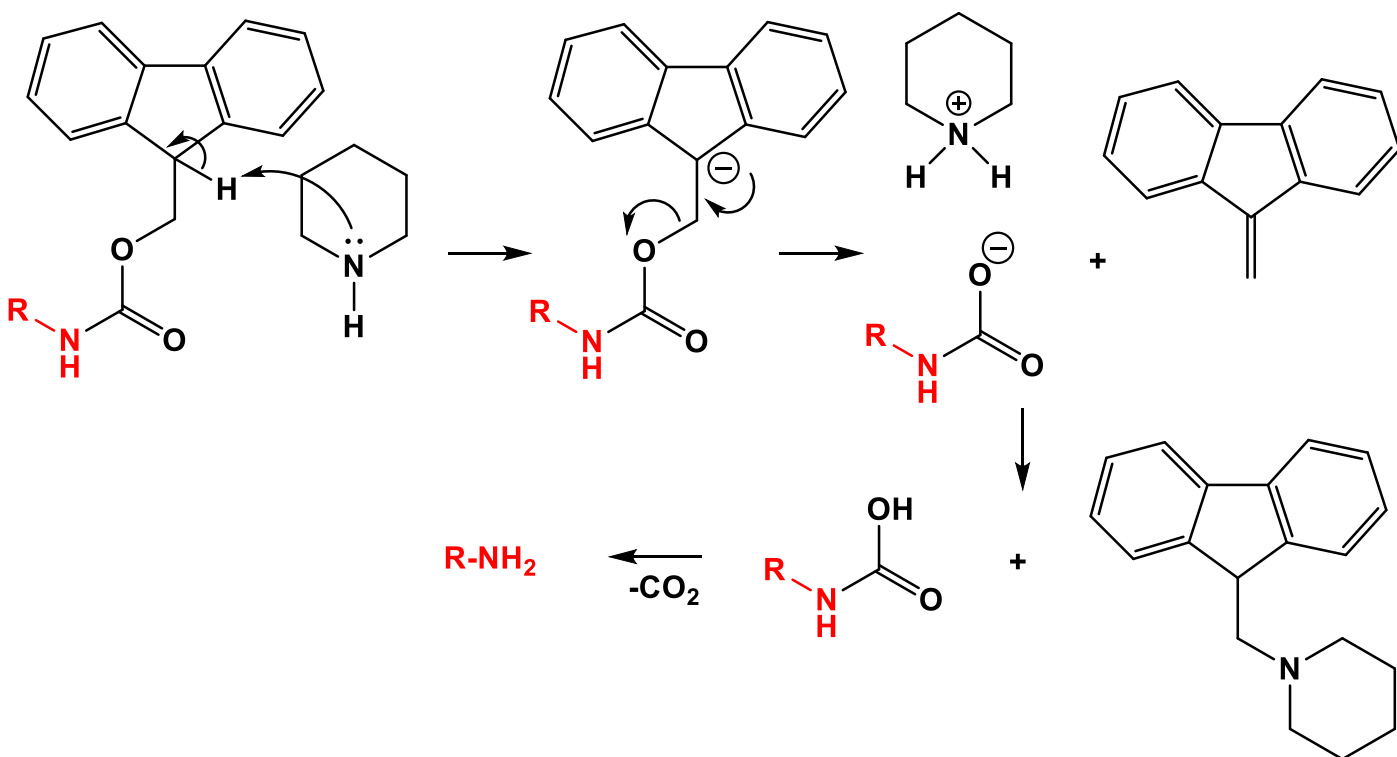


Chapter 3. Functional group protection

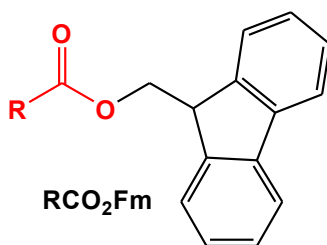
3.2.6. Protecting groups cleaved by β -elimination reactions

An important example of this subset is the 9-fluorenylmethoxycarbonyl (Fmoc) group.

It is commonly removed with secondary amines, such as piperidine, through a $E1cB$ mechanism. Piperidine removes the acidic proton at the C-9 position of the fluorene ring. This proton is acidic because the resulting anion is stabilized by the two aromatic rings. Dibenzofulvene is released, which reacts with piperidine forming a stable adduct that can be separated from the free amine.



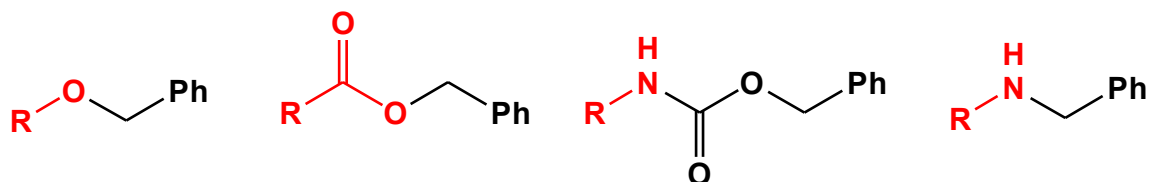
The 9-fluorenylmethyl (Fm) group is applied to the protection of carboxyl groups:



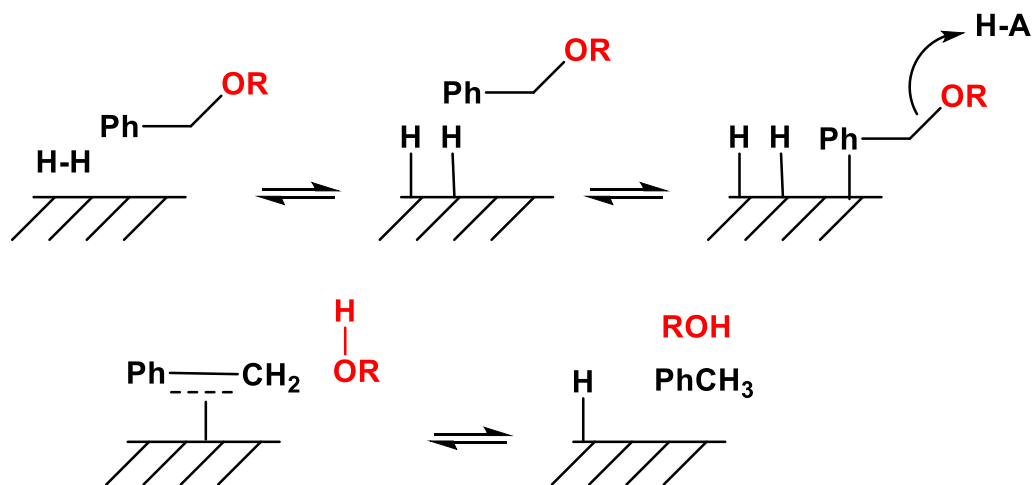
Chapter 3. Functional group protection

3.2.7. Protecting groups cleaved by hydrogenolysis

An excellent method for cleaving benzylic ethers, esters, carbamates, and amines



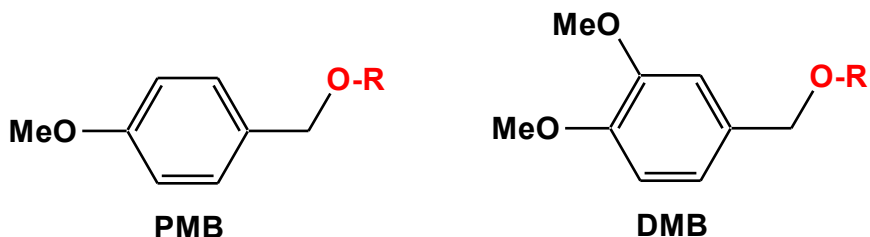
It uses H_2 in the presence of a transition metal such as Pd



Alternatively a process known as *catalytic transfer hydrogenation* can be employed. It uses 1,4-cyclohexadiene, cyclohexene, formic acid or ammonium formate as H_2 source.

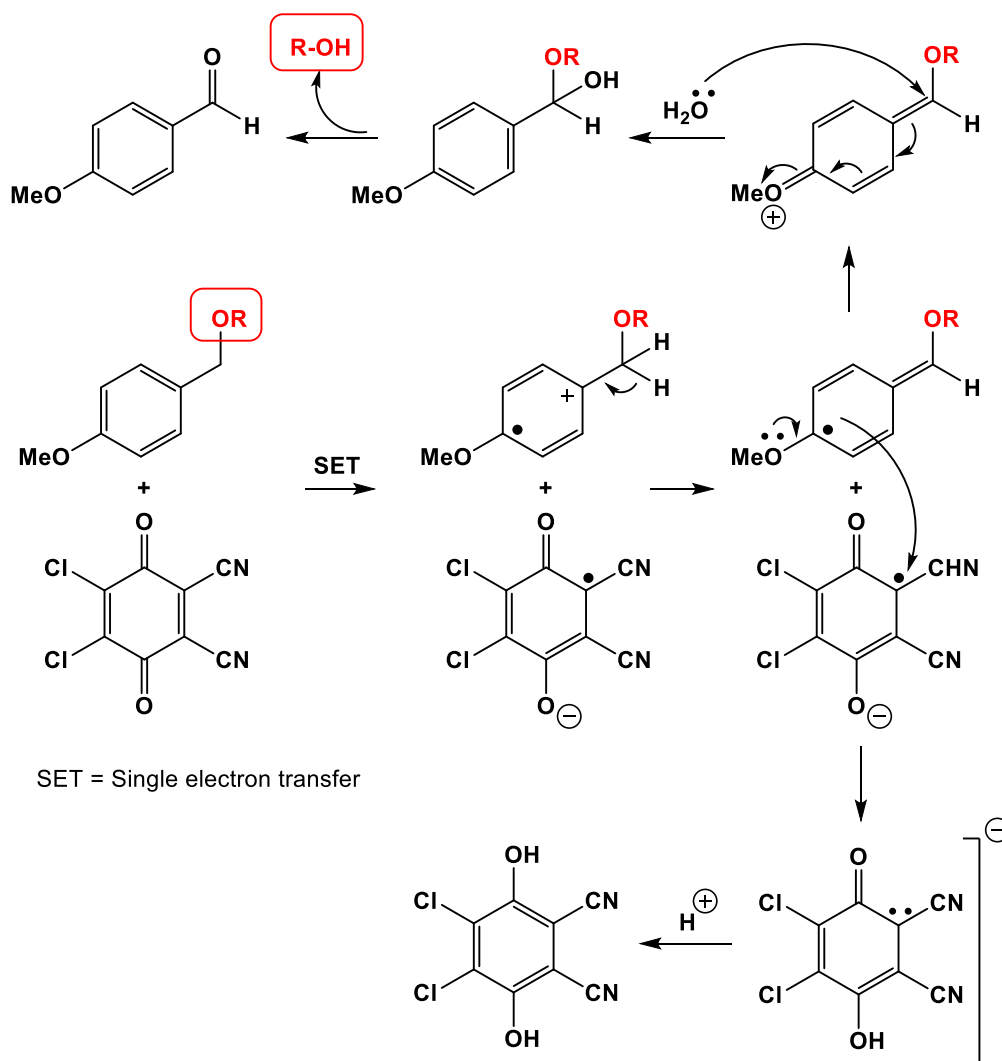
Chapter 3. Functional group protection

3.2.8. Protecting groups cleaved by oxidation



Oxidant reagents:

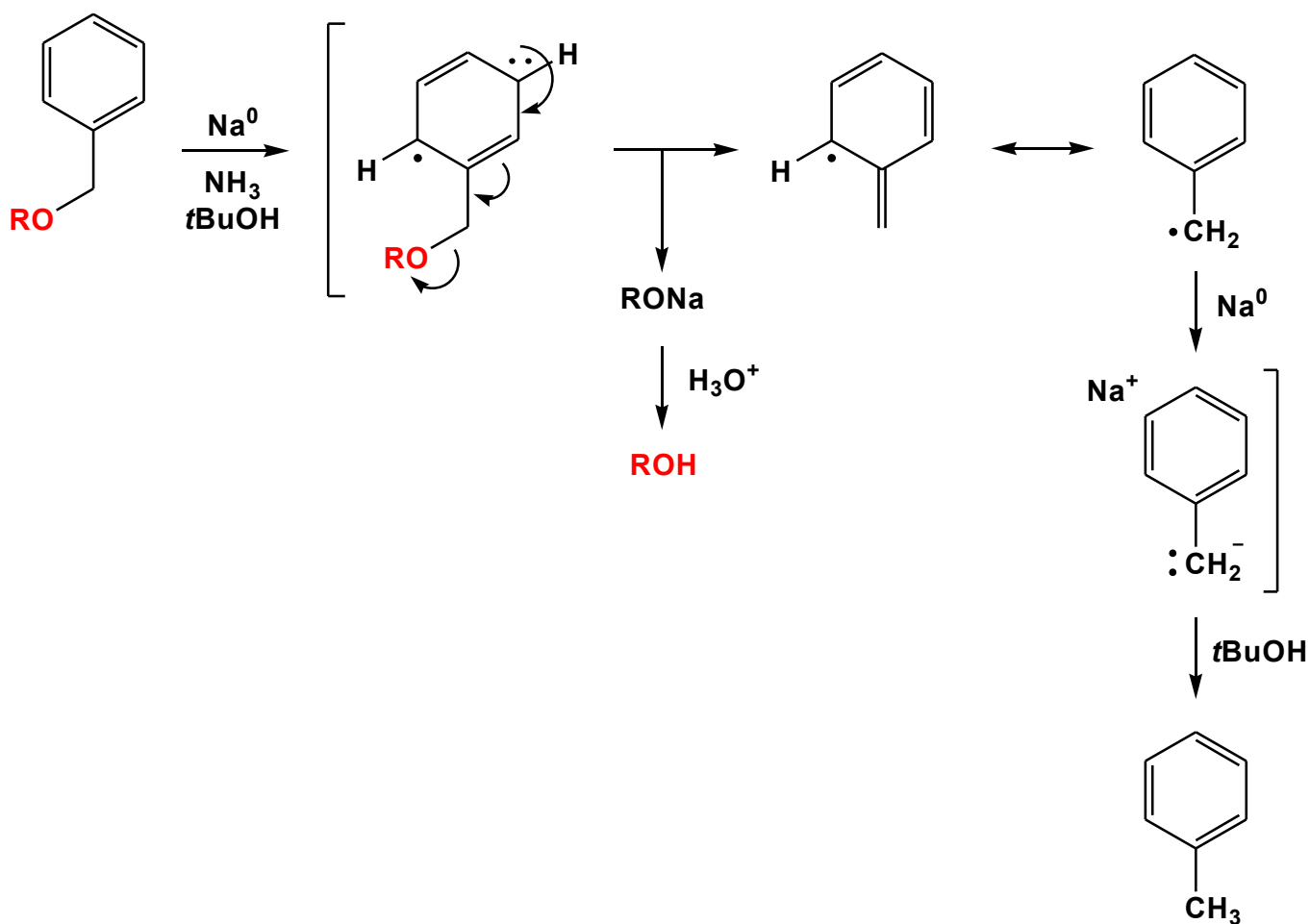
2,3-Dichloro-5,6-dicyano-1,4-benzoquinone (DDQ)
(NH₄)₂Ce(NO₃)₆ (CAN)



Chapter 3. Functional group protection

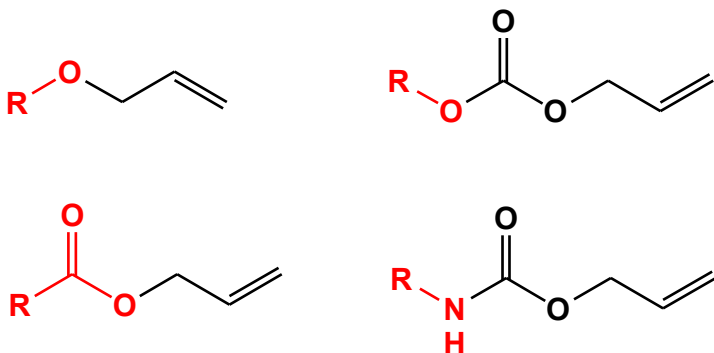
3.2.9. Protecting groups cleaved by dissolving metal reduction

Benzylic ethers and esters. Conditions: Na or Li in liquid NH_3

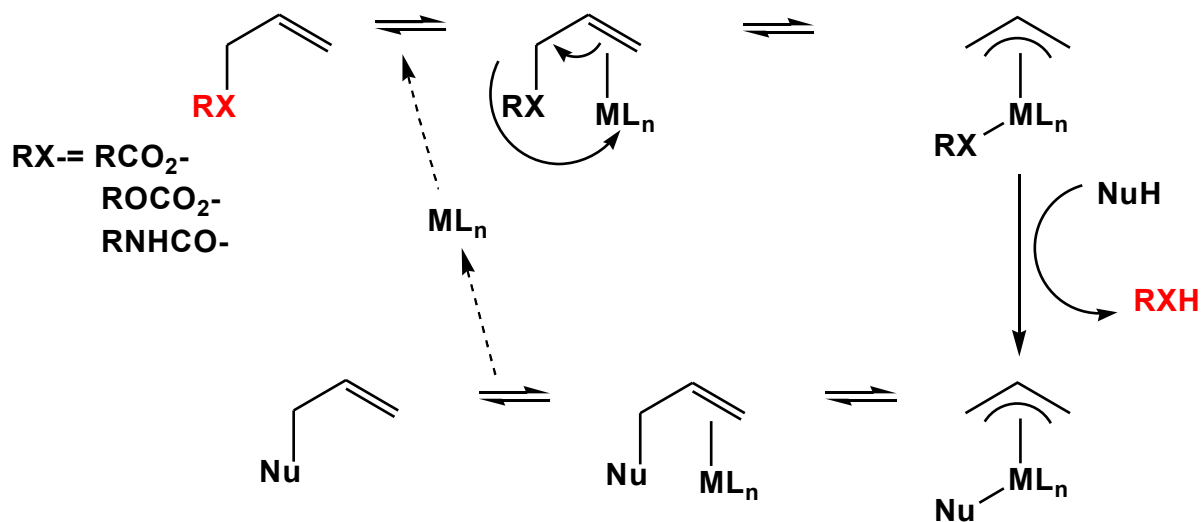


Chapter 3. Functional group protection

3.2.10. Protecting groups cleaved by transition metal catalysis



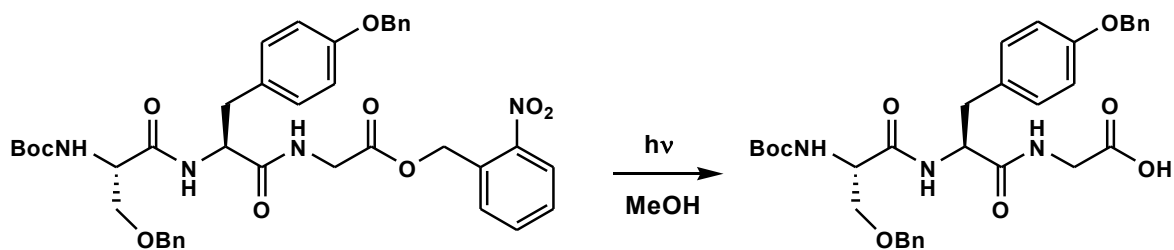
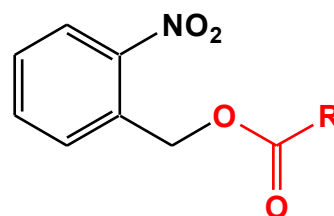
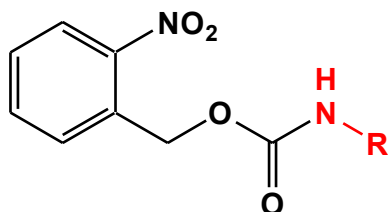
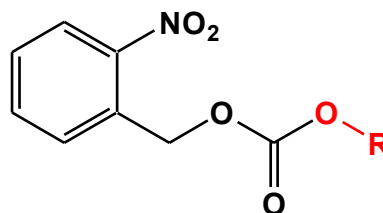
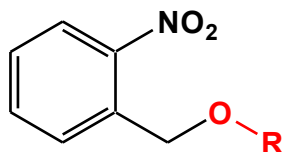
Conditions: $Pd(PPh_3)_4$, $PdCl_2$



NuH: Bu_3SnH , morpholine, AcOH

Chapter 3. Functional group protection

3.2.11. Protecting groups cleaved by light



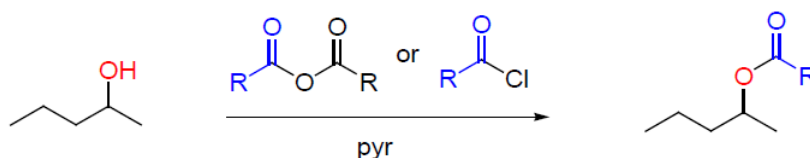
Chapter 3. Functional group protection

3.3. Functional group protection

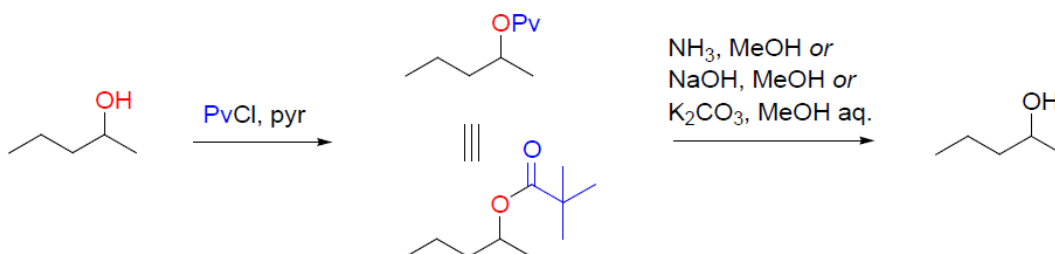
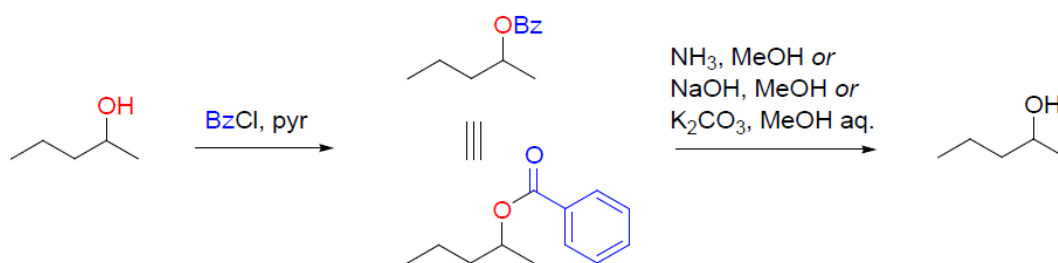
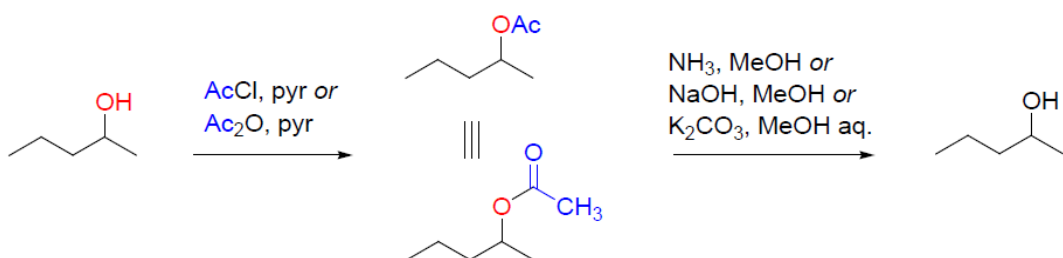
3.3.1. Hydroxyl protecting groups

Esters

Formation:



Examples and cleavage conditions:

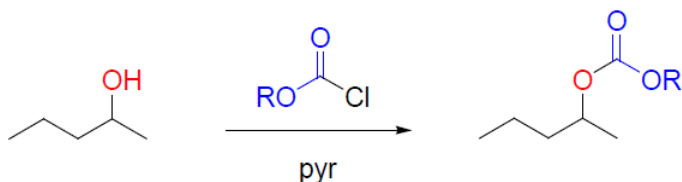


Reactivity order: Pv- < Bz- < Ac-

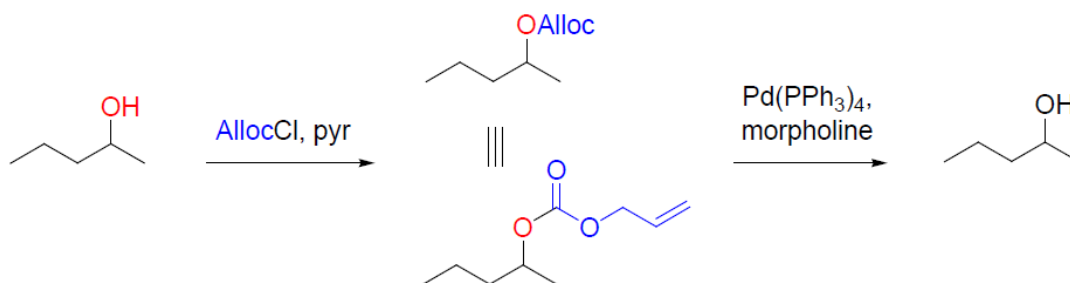
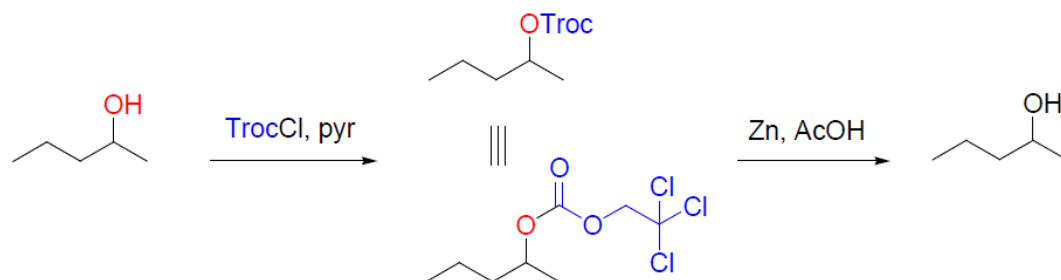
Chapter 3. Functional group protection

Carbonates

Formation:

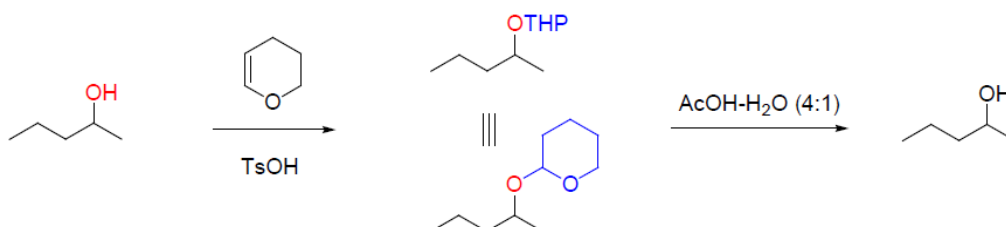


Examples and cleavage conditions:



Ethers

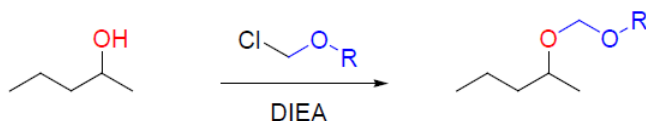
✓ Tetrahydropyranyl



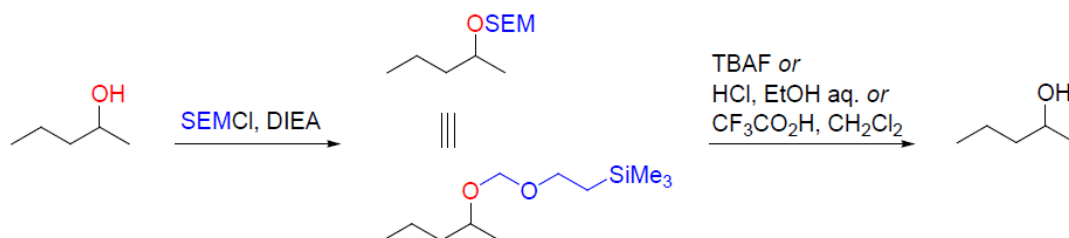
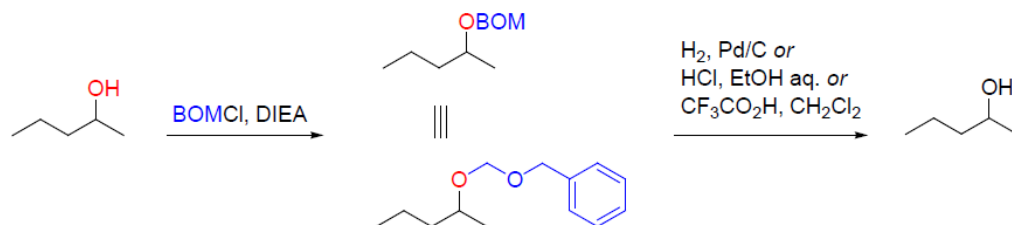
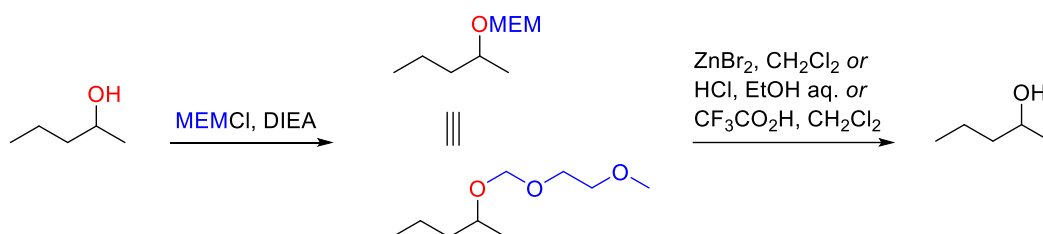
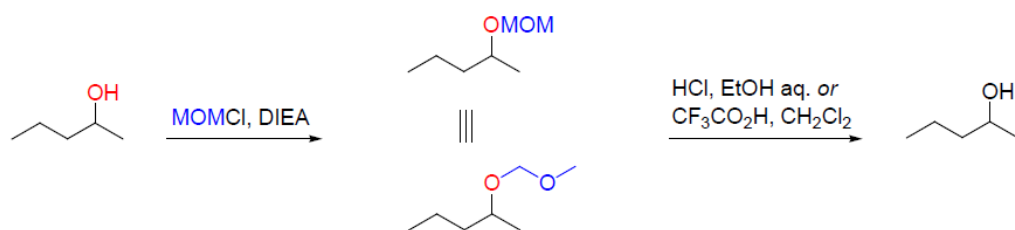
Chapter 3. Functional group protection

✓ Alkoxymethyl ethers

Formation:

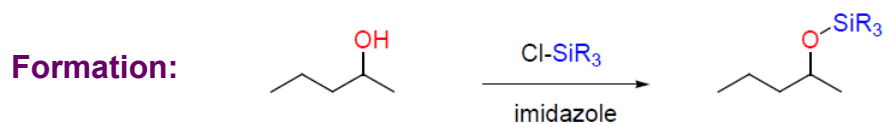


Examples and cleavage conditions:

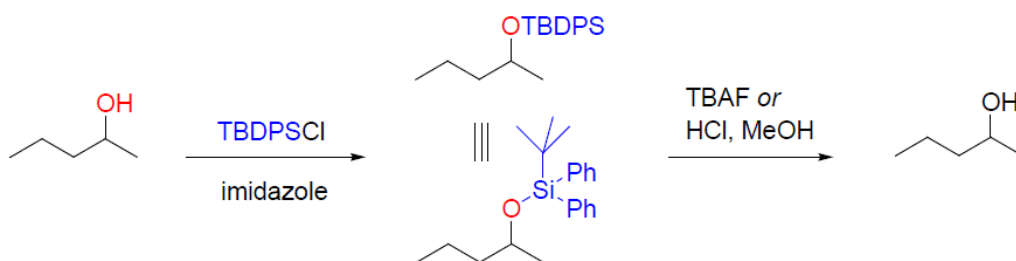
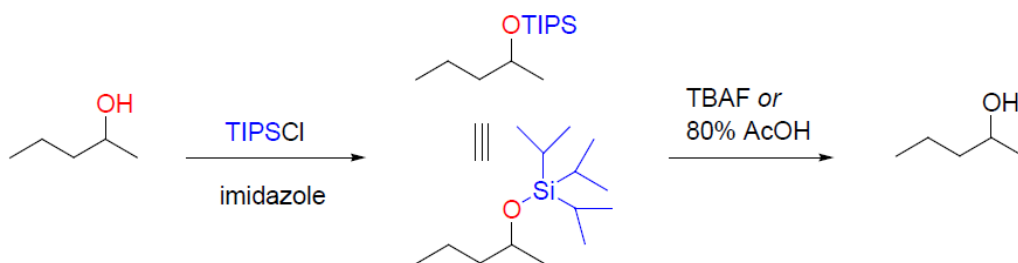
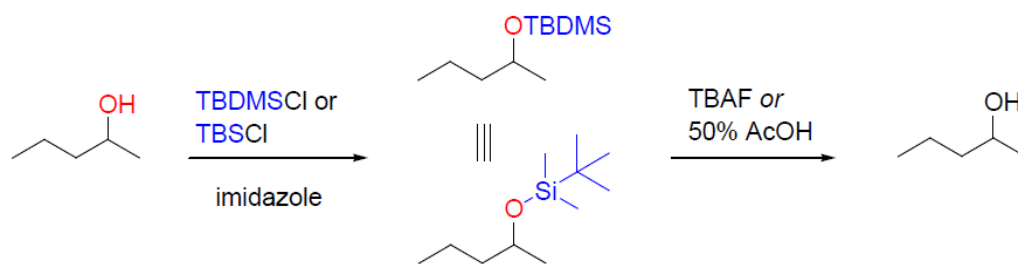


Chapter 3. Functional group protection

✓ Silyl ethers



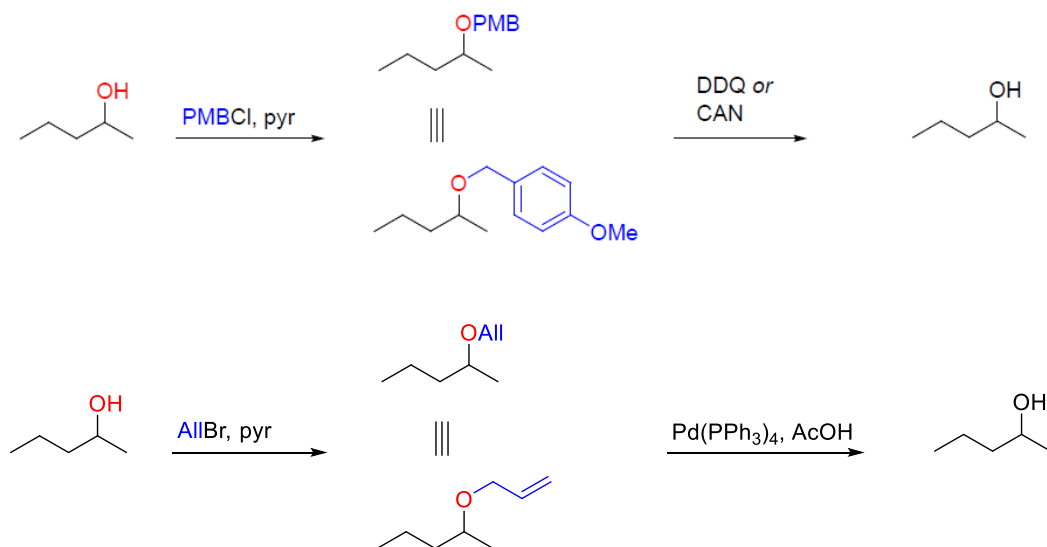
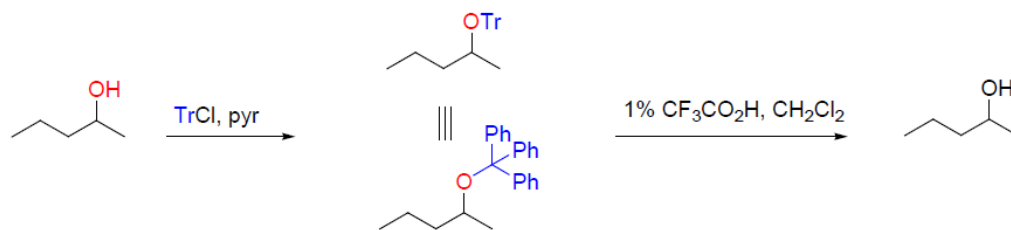
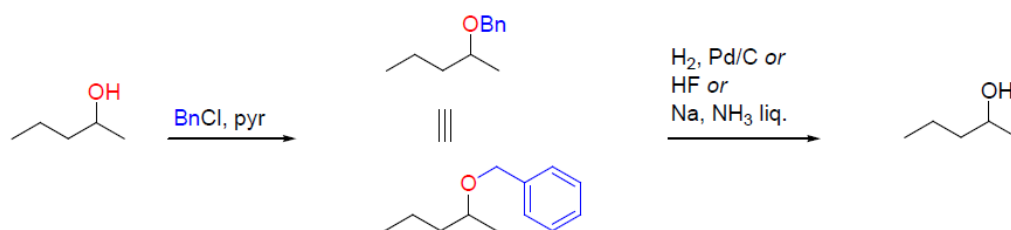
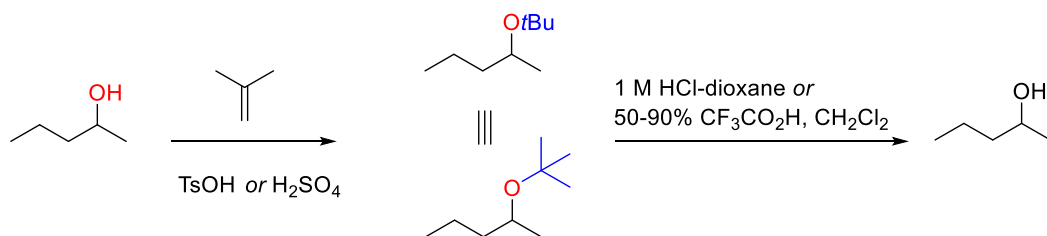
Examples and cleavage conditions:



Chapter 3. Functional group protection

✓ Alkyl ethers

Examples, formation and cleavage conditions:

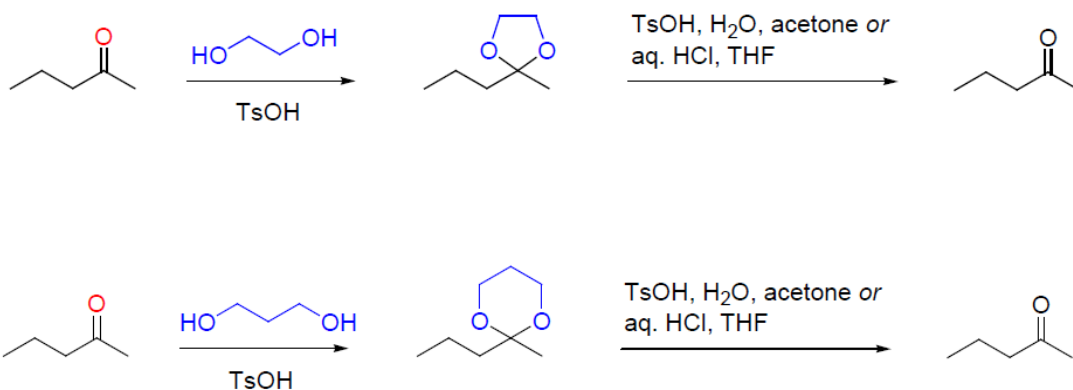


Chapter 3. Functional group protection

3.3.2. Carbonyl protecting groups

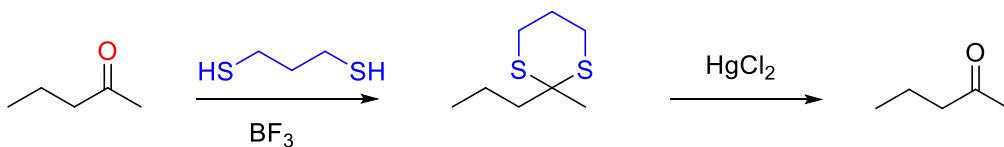
O,O-Acetals

Formation and cleavage conditions:



S,S-Acetals

Formation and cleavage conditions:

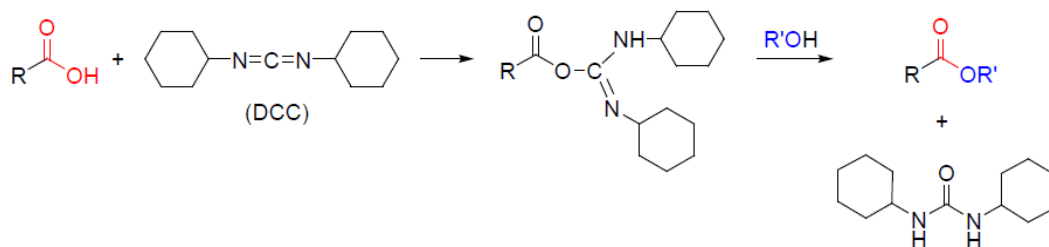
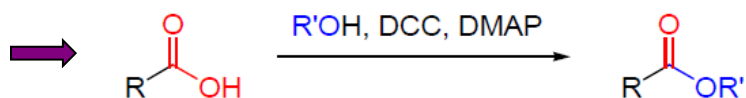
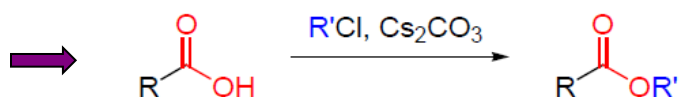
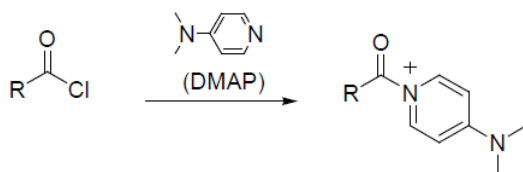
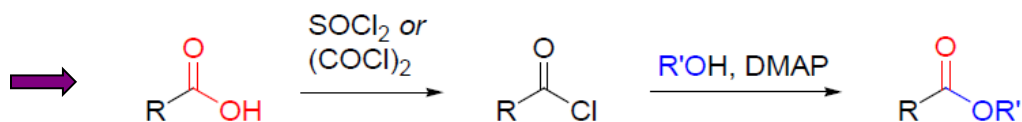
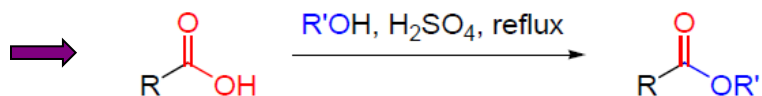


Chapter 3. Functional group protection

3.3.3. Carboxyl protecting groups

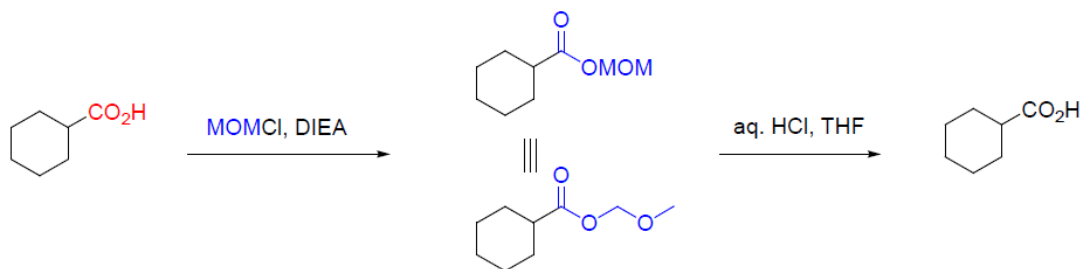
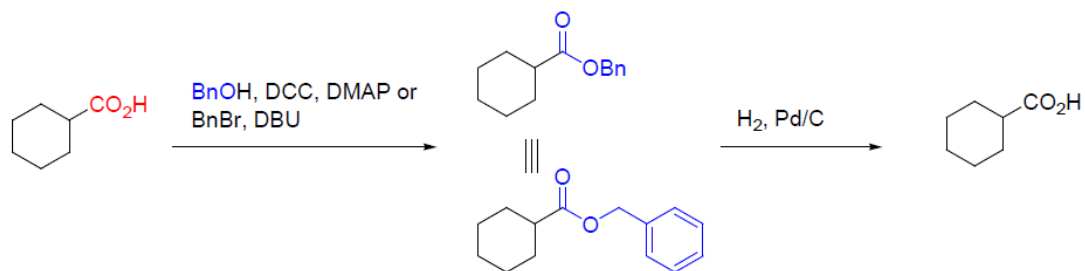
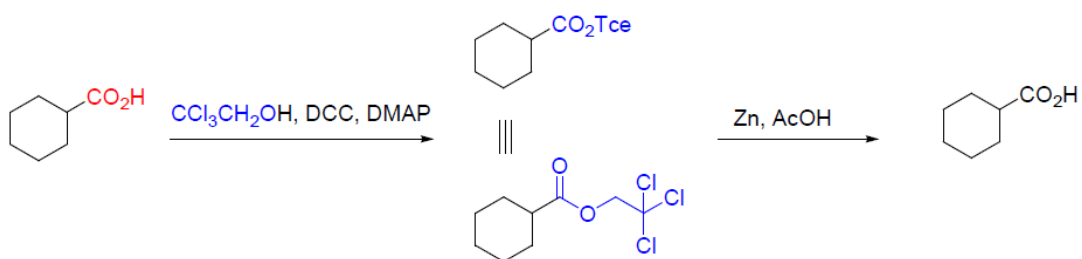
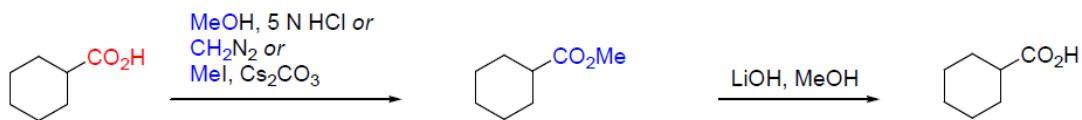
Esters

General esterification methods:

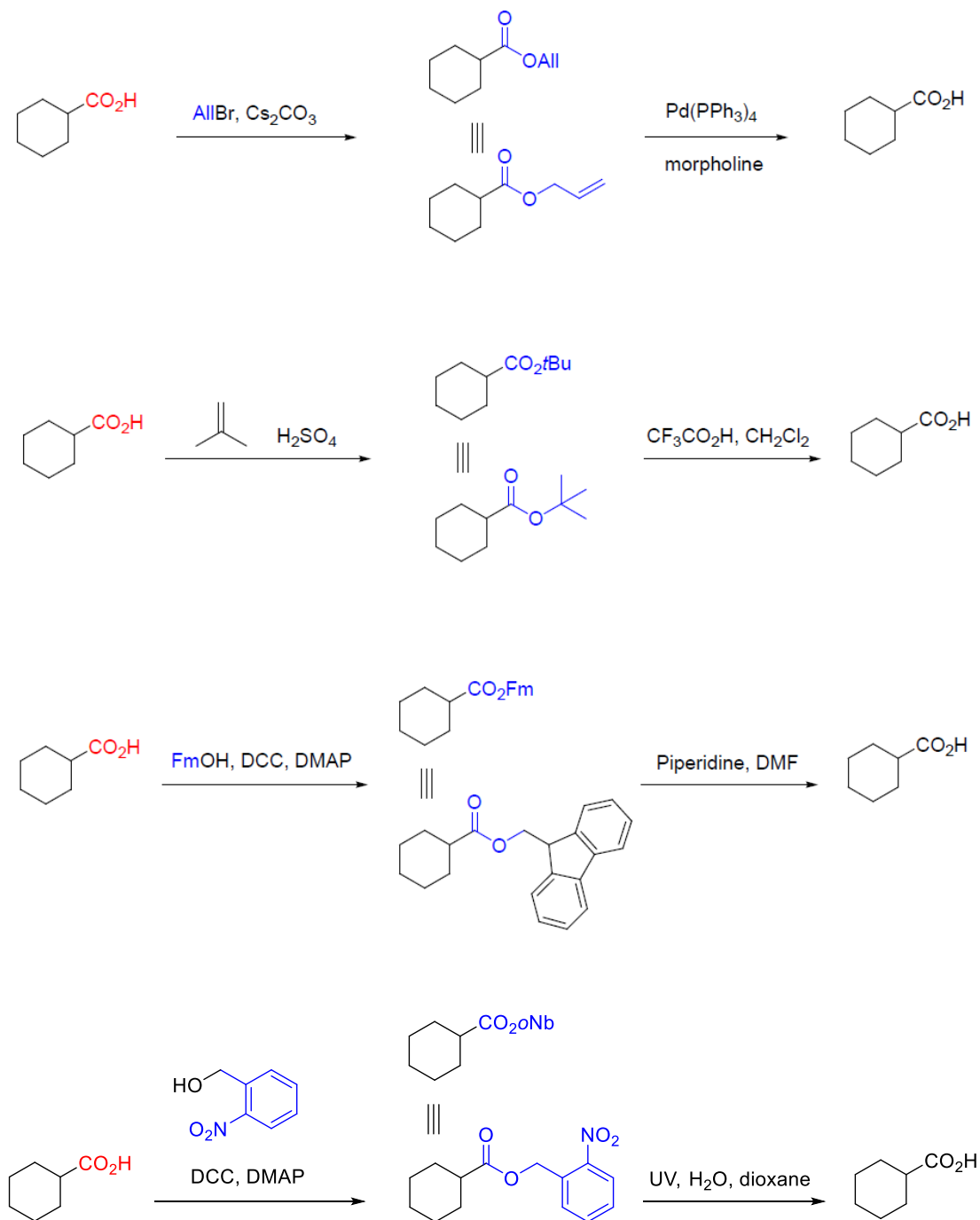


Chapter 3. Functional group protection

Examples, formation and cleavage conditions:



Chapter 3. Functional group protection



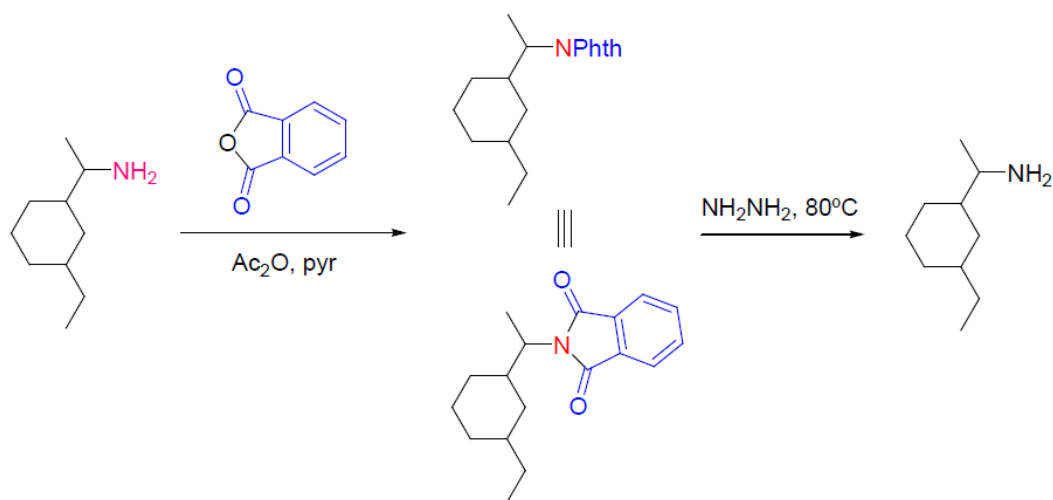
Chapter 3. Functional group protection

3.3.4. Amino protecting groups

Imides

Protecting groups for primary amines

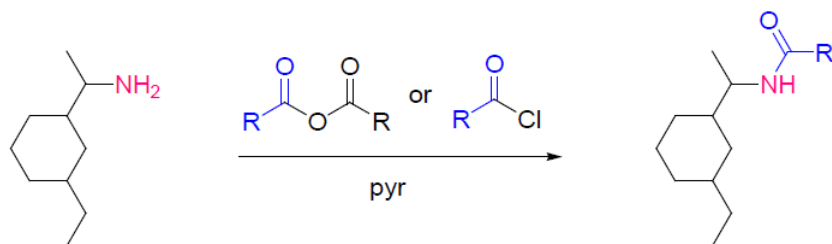
Formation and cleavage conditions:



Amides

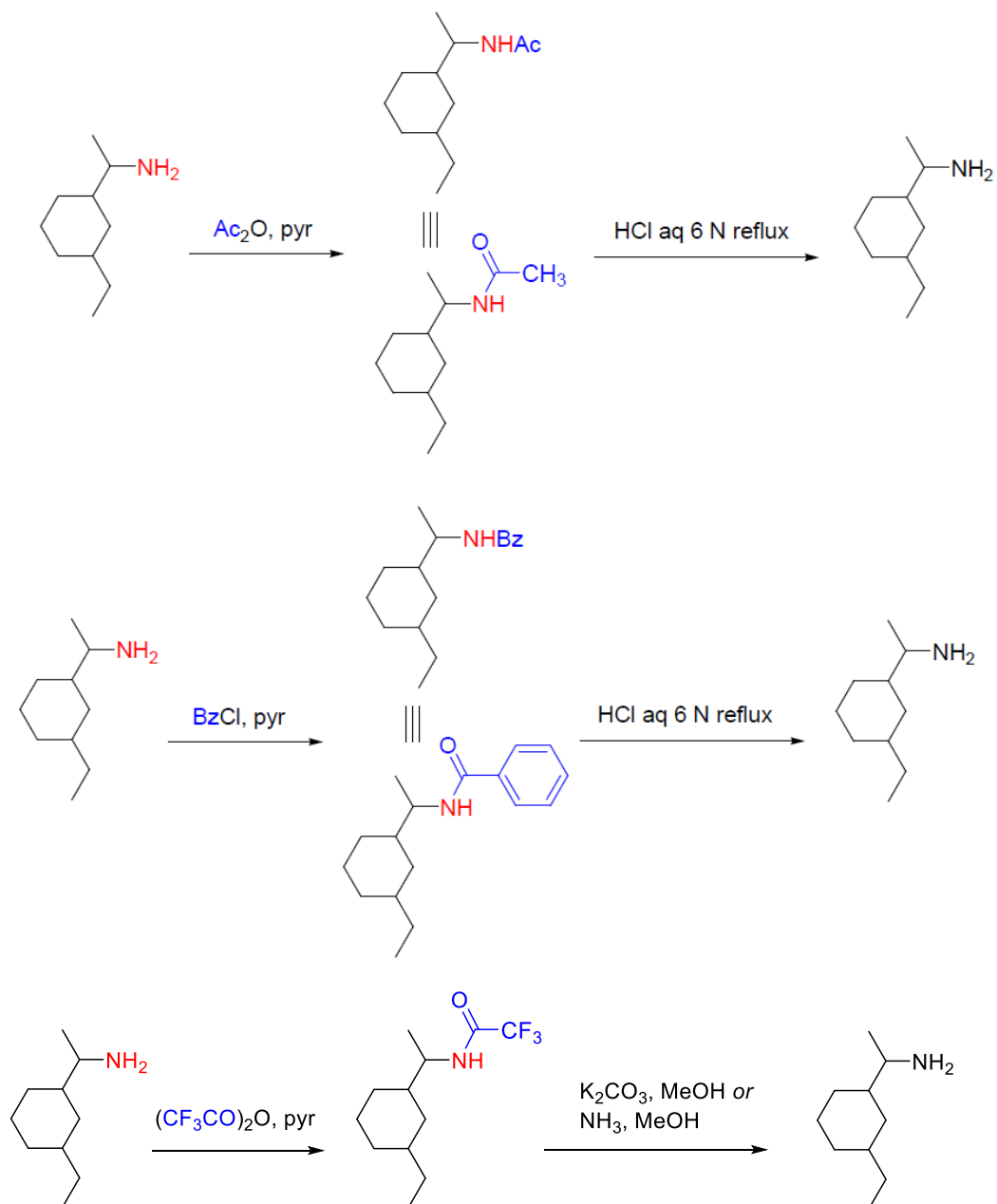
Protecting groups for primary and secondary amines

Formation:



Chapter 3. Functional group protection

Examples and cleavage conditions:

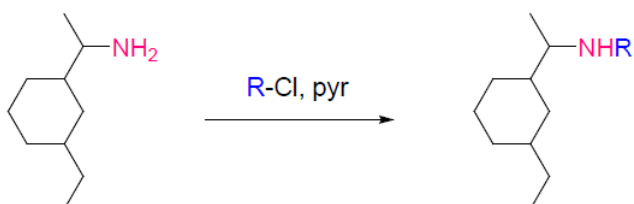


Chapter 3. Functional group protection

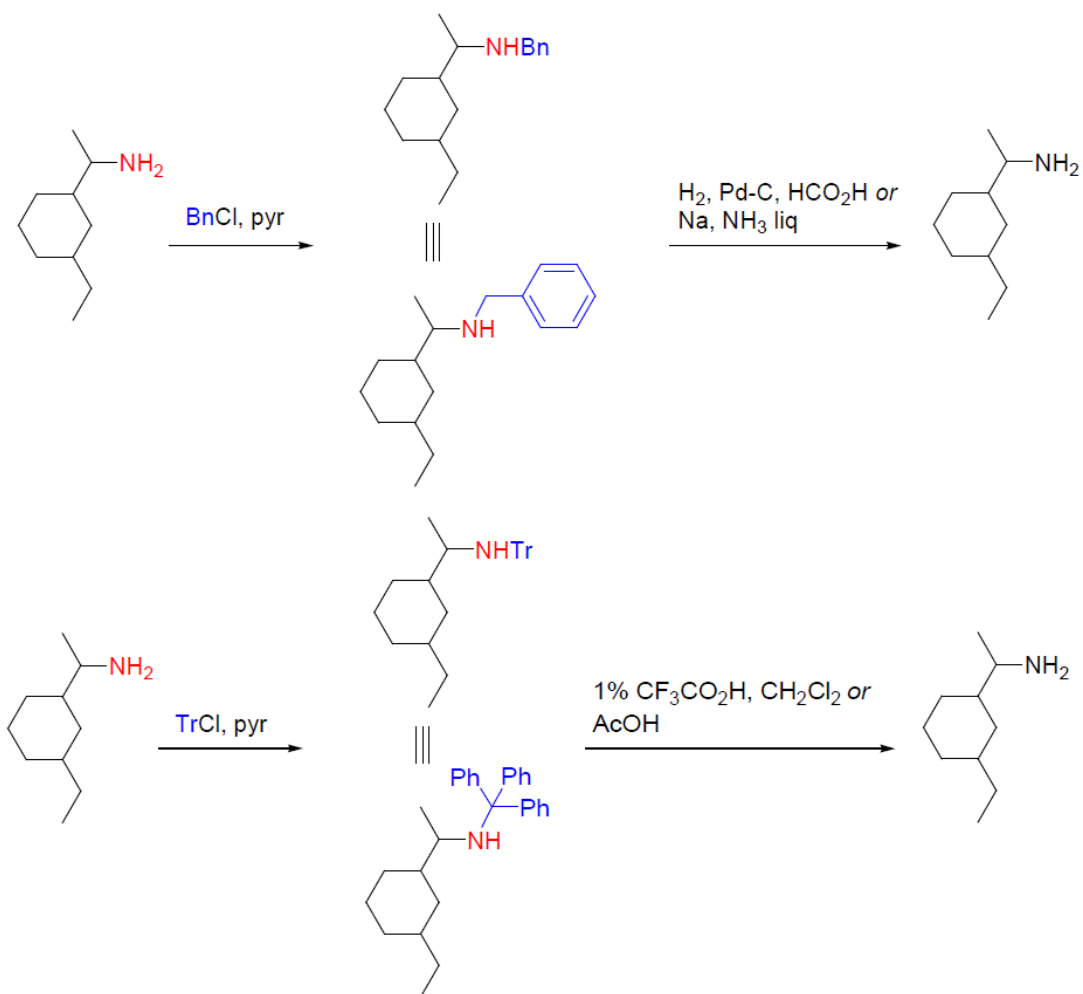
N-Alkyl derivatives

Protecting groups for primary and secondary amines

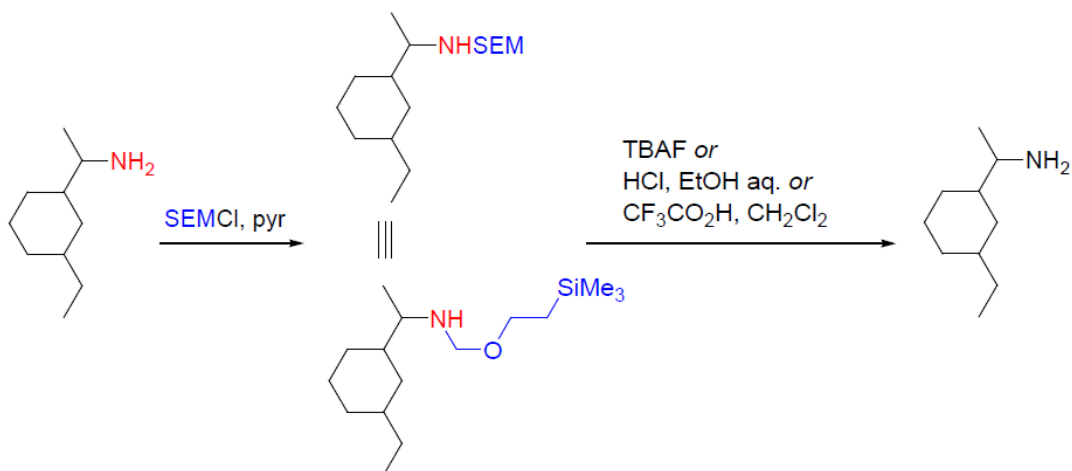
Formation:



Examples and cleavage conditions:



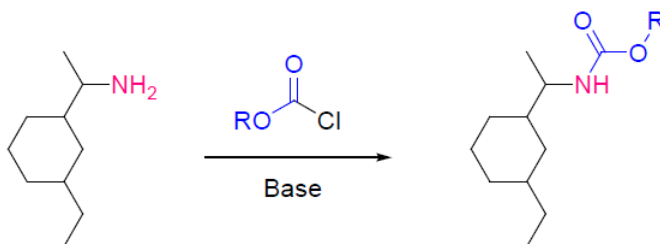
Chapter 3. Functional group protection



Carbamates

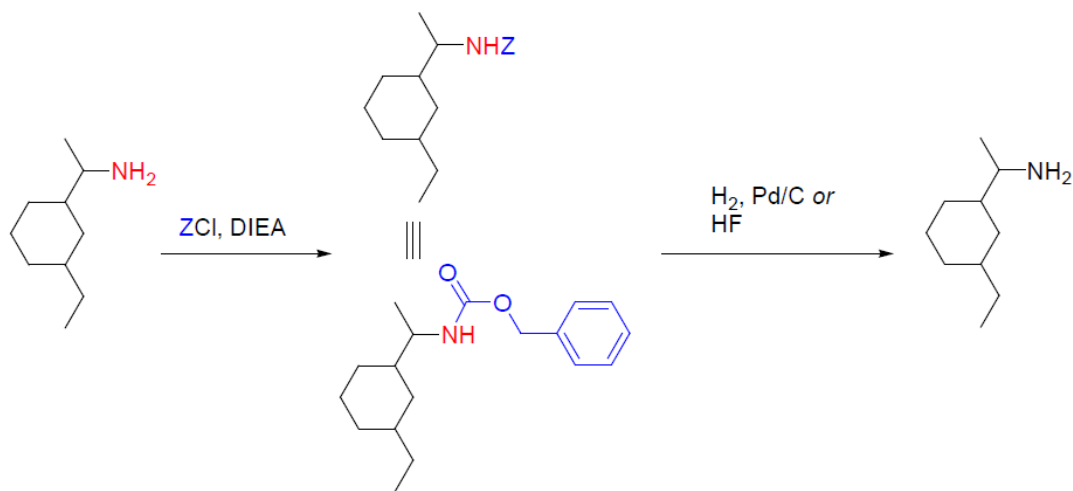
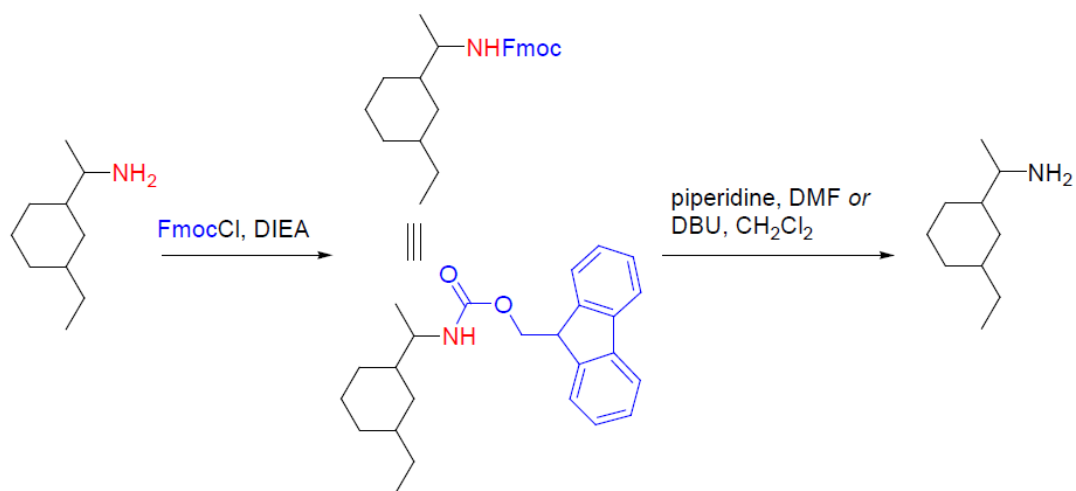
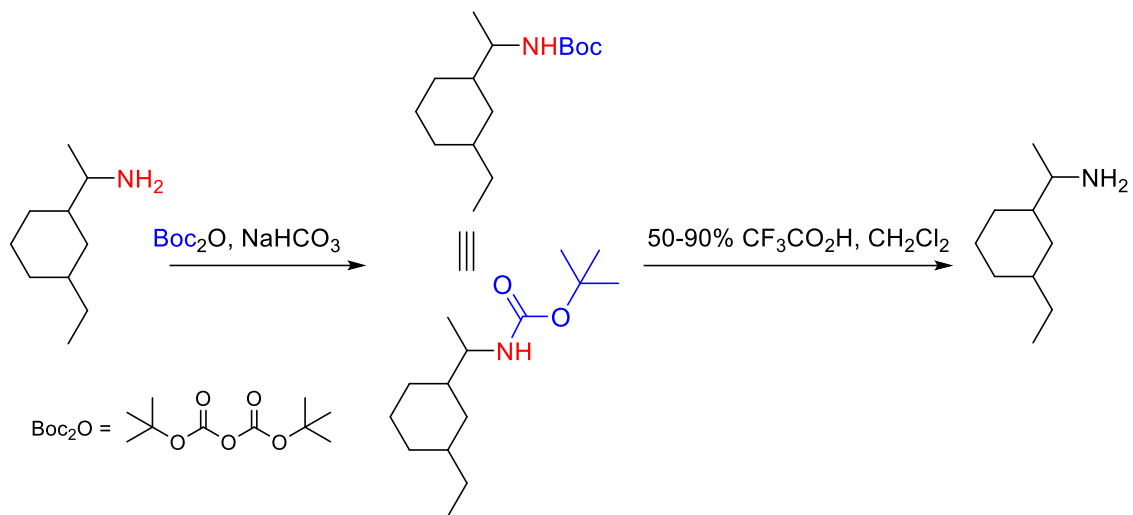
Protecting groups for primary and secondary amines

Formation:



Chapter 3. Functional group protection

Examples and cleavage conditions:



Chapter 3. Functional group protection

